Tan Yi Qi

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Education

Singapore University of Technology and Design (SUTD) **Singapore**
Bachelor of Engineering, Engineering Systems and Design (Minor in Psychology and Business Management) Sep 2024 – May 2028

VinUniversity **Hanoi, Vietnam**
Overseas Exchange — Innovation & Entrepreneurship 2025 Sep 2025 – Sep 2025
- Designated leader; managed group logistics and served as primary liaison for all trip activities
- Second Runner-Up, Business Investor Competition 2025

Singapore Polytechnic **Singapore**
Diploma in Applied Chemistry, specialization in Pharmaceutical Science Feb 2021 – Apr 2024

Relevant Projects

Web App & Power BI Dashboard with EcoPlus **SUTD**
Data & Business Analytics Project
- Built an integrated web app and Power BI dashboard for, Ecoplus, a piping/engineering firm to improve workforce visibility and manpower planning
- Developed order-entry and work-completion web forms that automatically updated job datasets in real time, replacing scattered manual records
- Designed a job status dashboard with colour-coded tracking (completed, overdue, due soon, not due) across order quantities, machine hours, and worker manhours
- Created Power BI visualizations covering machine utilisation/runtime by month, on-time vs. overdue job comparisons, and average manhours per worker

Skills: Data cleaning & consolidation, web app form development, Power BI, dashboard/UX design

Cost-Optimal Power Dispatch Under Carbon Constraints **SUTD**
Optimisation Project
- Extended a single-period Economic Load Dispatch (ELD) model into a multi-period cost-minimization framework accounting for Singapore's annual carbon emissions budget, generation mix, and import/solar constraints
- Formulated and solved a linear program across 17,520 half-hourly dispatch periods, subject to demand balance, generator bounds, N-1 spinning reserve, and solar/import limits
- Designed a "Debt-Aware Ratchet" Online Gradient Descent (OGD) algorithm to adaptively update a carbon penalty in response to real-time emissions deviations from a budget glide path
- Achieved 33.72 Mt CO_2 actual emissions against a 33.73 Mt budget (0.03% under), while quantifying the cost premium of carbon-constrained dispatch versus market price

Skills: Linear programming/optimization, Python, adaptive control (online gradient descent), energy systems modelling

Aggregate Planning Analysis with SATS / Country Foods **SUTD**
Manufacturing & Service Operations Project
- Mapped the end-to-end overseas food supply chain (raw material sourcing $\rightarrow$ overseas kitchens $\rightarrow$ shipping $\rightarrow$ SG warehouse $\rightarrow$ uplift) with a 4–7 month total lead time, identifying key production, MOQ, and warehouse capacity constraints
- Built a baseline chase-strategy aggregate planning model in Excel across chicken, fish, and seafood product lines, estimating total operating cost at $32.5M
- Developed a Python-based LP optimization model to minimize combined production, outsourcing, and holding costs across the full planning horizon, improving on the baseline to $32.2M
- Benchmarked SATS's real-world SKU-level MPS/MRP planning process against theoretical family-level aggregate planning, evaluating trade-offs in forecast smoothing, precision, and planning maturity

Skills: Aggregate planning, Excel modelling, Python optimization, supply chain mapping, S&OP evaluation

Recycling Collection Routing Problem **SUTD**

Heuristics and Systems Theory Project

- Modeled Singapore's HDB recycling truck collection as a Multi-Depot, Multi-Trip Vehicle Routing Problem with capacity, shift-duration, and depot-unloading constraints
- Sourced and geocoded 998 real HDB recycling bin locations via Singapore's OneMap API, building distance/time matrices from haversine distances with road-detour scaling
- Formulated a MILP with MTZ subtour elimination and solved it exactly using Gurobi on small instances, benchmarking against Simulated Annealing and Genetic Algorithm metaheuristics
- Simulated Annealing outperformed Gurobi's exact solution by 2.3$\times$ (93.22 vs. 214.14 objective) in 14.7 seconds vs. 6 hours on a 30-customer benchmark, demonstrating the practicality of metaheuristics at scale
- Solved the full 574-customer real-world instance with 100% coverage, where exact methods became computationally infeasible

Skills: MILP formulation, Gurobi, metaheuristics (Simulated Annealing, Genetic Algorithm), Python, geospatial data processing (OneMap API)

Work Experience

National Cancer Center Singapore (NCCS Business Office) **Apr 2024 – Jul 2024**

Temporary Senior Associate Singapore

- Handled billing under eligible government healthcare schemes to ensure accurate and timely invoicing over a continuous 6-week period; verified patient treatment details
- Developed step-by-step billing procedure guidelines and supported training sessions to ensure consistency and accuracy in patient billing operations

Inspectorate (Bureau Veritas) **Mar 2023 – Aug 2023**

Laboratory Intern Singapore

- Prepared and standardized chemical reagents following safety protocols and experimental specifications, and calibrated instruments over a 6-month period
- Calibrated machinery and instruments to ensure compliance with universal and international standards on a monthly basis
- Configured laboratory setup and organization to streamline workflow and compliance with safety standards
- Coordinated cross-departmental collaboration to facilitate a seamless laboratory relocation, ensuring minimal disruption to operations over a 3-week relocation period

Extracurricular Activities

Development and Dice Club, SUTD **Mar 2025 – Mar 2026**

Secretary

- Handled administrative work and finalized documents
- Organized and led club events and activities, including Game Jam 2025 (annual competitive digital and physical game-making event)

Class Executive Committee, SUTD **Sep 2024 – Apr 2025**

Sustainability and Safety Advisor

- Instilled sustainability values and actions in the workspace
- Ensured safety throughout class activities

Technical Skills

Programming Languages: Python, R, SQL, Julia
Software: Microsoft Office (Word, Excel), Google Workspace (Sheets, Apps Script, Power BI), Adobe (Acrobat, Photoshop)
Other: Laboratory & Fabrication, Blender & Rhino